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Blood Flow Restricted Electrical Stimulations to Prevent or Attenuate Symptoms of Muscle Damage

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ABSTRACT

The objective of this study was to determine if performing electrical stimulations (E-STIM) under blood flow restriction (BFR) would result in a greater protective effect against symptoms of muscle damage. 18 individuals (9 females) completed a damaging bout of exercise followed by a low frequency E-STIM treatment protocol on both arms, one of which was completed under BFR. The treatment protocol was then repeated 24-hours post-exercise. There were main effects of time for muscle thickness (pre: 3.5 cm; 48 h post: 3.8 cm; $BF_{10} = 88.476$), discomfort (pre: 0.0 au; 48 h post: 4.2 au; $BF_{10} = 241.996$), and isometric strength (pre: 278 N; 48 h post: 232 N; $BF_{10} = 10{,}289.894$) which all changed as a result of the damaging exercise protocol, but there were no differences between conditions [all Bayes Factors (BF_{10}) < 0.28]. The effectiveness of low frequency E-STIM for preventing the onset of exercise-induced muscle damage would not appear to be enhanced if performed under BFR.

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KEYWORDS
DOMS; E-STIM; occlusion; recovery; soreness; swelling

Introduction

Novel eccentric exercise is known to induce muscle damage which results in prolonged decrements in force production, limited range of motion, muscle swelling, delayed onset muscle soreness (DOMS), and elevated blood enzyme levels (Clarkson & Hubal, 2002). Methods aimed at preventing or treating muscle damage are desirable for many reasons such as increasing adherence to exercise, and allowing athletes to recover and maximize force output during a subsequent event. One of the most effective methods for alleviating DOMS involves pharmacological treatment with non-steroidal anti-inflammatory drugs (NSAIDs). While effective, NSAIDs increase the risk of gastrointestinal and cardiovascular complications (Davis & Robson, 2016) and can even blunt the adaptive response to exercise (Howatson & van Someren, 2008). Alternate means of treating DOMS include exercise (Isabell et al., 1992), cryotherapy (Crystal et al., 2013), compression (Kraemer et al., 2001), and electrical stimulations (E-STIM) (Babault et al., 2011), but the effects of these modalities are usually transitory and are administered after muscle damage has already presented itself (Cheung et al., 2003).

In addition to examining potential treatments of muscle damage, researchers have also examined methods to prevent the onset of muscle damage (Vanderthommen et al., 2007). One approach involves the use of low frequency E-STIM (i.e. <10Hz) to induce visual but

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comfortable muscle contractions (Babault et al., 2011). When applied 10-min after damaging knee extensor exercises, low frequency E-STIM was unable to limit force loss or discomfort, but did lower creatine kinase levels 3-days post-exercise implying a reduction in damage to the sarcolemma (Vanderthommen et al., 2007). The authors hypothesized this may be the result of increased blood flow to the muscle which may have reduced the inflammatory response (Vanderthommen et al., 2007). While the results on E-STIM have been inconclusive, it appears to be the most effective for limiting sensations of soreness (Babault et al., 2011) and may be enhanced if combined with another treatment modality.

In addition to the potential of E-STIM to reduce the magnitude of exercise-induced muscle damage, the application of blood flow restriction (BFR) has also shown some promise. The use of BFR is primarily studied to acutely induce a hypoalgaesic effect (Song et al., 2021), but other studies have suggested that it could potentially prevent the onset of muscle damage (Page et al., 2017). For instance, repeated bouts of inflations and deflations in the absence of exercise were shown to limit force loss, creatine kinase levels, and sensations of discomfort when administered after damaging exercise (Page et al., 2017). Furthermore, repeated bouts of inflations and deflations have been shown to augment the recovery of strength and muscle activation for those recovering from strokes (Hyngstrom et al., 2018). The enhanced recovery of strength resulting from BFR was hypothesized to be, in part, related to increased group III/IV afferent activity and an increased excitability of motor neurons (Hyngstrom et al., 2018). While these studies suggest a potential role of BFR to limit muscle damage and enhance recovery, conflicting results also show that the application of BFR combined with unloaded elbow flexion exercises was not effective at limiting exercise induced muscle damage (Wong et al., 2020).

The combined use of BFR and E-STIM in the literature is scarce. Acute studies have examined the effects of applying BFR during a single bout of E-STIM to see how it impacts muscle fatigue (Cole & Brown, 2000; Head et al., 2020), metabolism (Cohen et al., 2022; Greenhaff et al., 1993), and hormone concentrations (Inagaki et al., 2011). Additionally, chronic training interventions have examined whether the addition of BFR can enhance the effectiveness of E-STIM for increasing vascular function (Cohen et al., 2021; Gorgey et al., 2016) and muscular adaptations (Gorgey et al., 2016; Natsume et al., 2015; Slysz & Burr, 2018; Slysz et al., 2020) when performed across numerous training sessions. Nonetheless, no studies, to our knowledge, have examined the efficacy of blood flow restricted electrical stimulations as it relates to preventing or attenuating the onset of symptoms associated with exercise induced muscle damage. Given the effectiveness of BFR at enhancing hypoalgesia when combined with low load exercise (Song et al., 2021), it is reasonable to hypothesize that the effectiveness of E-STIM may be augmented if performed under BFR. Therefore, the purpose of the present study was to test whether the application of BFR during low-frequency E-STIM could help prevent exercise-induced muscle damage.

Materials & methods

Participants

To be eligible for participation in the study all individuals were required to be between the ages of 18 and 65, they could not have an implanted electrical device, could not perform upper body exercise starting 48 hours before their initial visit up until completion

of the study, and they could not have a systolic blood pressure of ≥140 mm Hg or diastolic blood pressure of ≥90 mm Hg. Individuals were also excluded from the study if they had more than one risk factor for thromboembolism which included: (1) body mass index ≥30; (2) diagnosed with Crohn's or inflammatory bowel disease; (3) a past fracture of the hip, pelvis, or femur; (4) any major surgeries in the last 6 months; (5) diagnosed with varicose veins; (6) a family history of either deep vein thrombosis or pulmonary embolism; (7) taking oral contraceptives; or (8) use of tobacco. This study was approved by the University's institutional review board, and all individuals provided informed written consent before participating.

Study overview

The study involved a total of 3 visits. On visit 1, the participants were screened for inclusion in the study and then provided baseline ratings of discomfort in each of their arms before having their arterial occlusion pressure measured. Afterwards, participants had their muscle thickness measured via ultrasound and completed isometric and 1RM strength testing. After the 1RM strength testing, individuals performed a bout of damaging eccentric exercise on both arms. One arm was assigned to complete the E-STIM intervention while the other arm completed the same E-STIM intervention while also having BFR applied (E-STIM + BFR). The treatment conditions were counter-balanced for limb dominance to ensure an equal number of individuals completed each treatment condition on their dominant arm. Immediately after the damaging exercise, individuals completed the E-STIM and E-STIM+BFR treatments on both limbs to conclude visit 1. Visit 2 took place 24 h after visit 1 and involved ratings of discomfort, muscle thickness, and isometric strength. Afterwards, individuals then completed the treatment protocol a second time. Visit 3 took place 24 h after visit 2 and consisted of ratings of discomfort, muscle thickness, and isometric strength. A visualization of the study overview is shown in Figure 1.

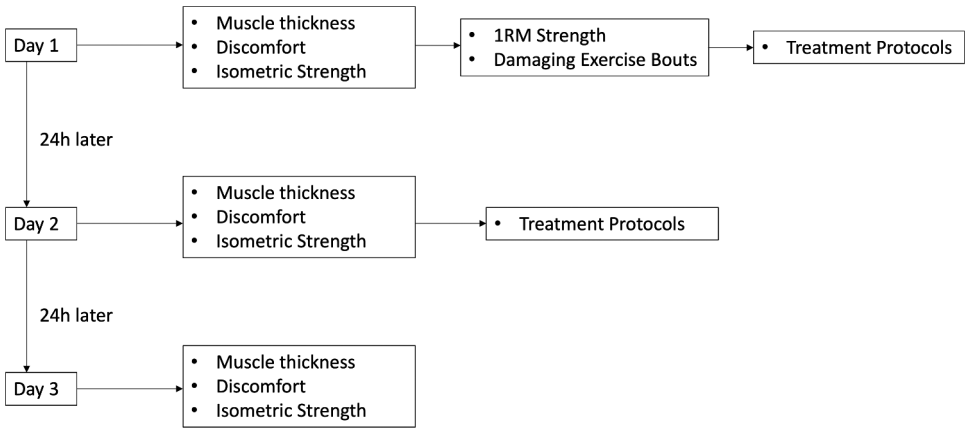

Figure 1. Study overview.

Arterial occlusion pressure

With the individual standing, a 5 cm wide nylon cuff (Hokanson inc.; Bellevue, WA; USA) was placed at the most proximal portion of the arm, and a doppler probe (Arjo inc.; Addison, IL; USA) was placed at the radial artery at the wrist. The pressure in the cuff was increased in 1 mmHg increments until an auditory sound of the individual's pulse could no longer be detected. The lowest pressure at which a pulse could no longer be detected was recorded as the individual's arterial occlusion pressure. This measurement was performed on the arm assigned to complete the E-STIM+BFR protocol and was performed to ensure relative pressures were administered (Loenneke et al., 2013).

One-repetition maximum (1RM) test

Maximum concentric strength (the heaviest weight that could be lifted for a single repetition) of the participants' arms were tested using the unilateral elbow flexion exercise completed with dumbbells. This measurement was used to set the workload for the exercise bouts. The participant completed the same protocol on both arms. All 1RM attempts were separated by 90 seconds of rest and were performed with the individuals back and heels against a wall to ensure strict form. All 1RMs were measured to the nearest 0.2 kg and were usually obtained in within 5 attempts.

Damaging exercise protocol

Individuals stood with their heels and back against a wall to ensure strict form. Eccentric elbow flexion exercises were performed to volitional failure using a load corresponding to 80% of the individuals concentric 1RM. The participants were handed a dumbbell with their elbow fully flexed and were required to lower the dumbbell slowly and in a controlled fashion to the beat of a metronome so the eccentric muscle action took 2s to complete. The load was then taken from the participant when their elbow was fully extended and the participant completely flexed their elbow before having the load handed back to them. Failure was defined as the point at which the participant could no longer lower the load in a controlled fashion to the beat of the metronome. A 5-min rest period was given before completing the damaging exercise on the contralateral arm.

Treatment protocols

The E-STIM protocol was administered consistently across both arms using a table top E-STIM device (Intellect Legend XT, Dallas, TX, USA). Two electrodes (5 cm × 5 cm) were placed at each the proximal and distal portion of the primary elbow flexor muscles. Symmetrical biphasic rectangular pulses lasting 200µ at a frequency of 5 Hz were administered to develop a continuous and visible twitch over a 20-minute period. This protocol is similar to what has been employed previously in attempt to enhance recovery from exercise-induced muscle damage (Babault et al., 2011). During the E-STIM treatment protocols, a 5 cm wide nylon cuff was placed at the most proximal portion of each limb. The E-STIM arm did not have the cuff inflated for the 20-minute duration. The E-STIM+BFR arm had the cuff inflated to 100% of the individual's arterial occlusion pressure for

2-minute bouts each separated by a 1-minute rest period as has been used previously in conjunction with E-STIM (Slysz & Burr, 2018; Slysz et al., 2020). This continued until the 20-minute duration was completed. Thus, both arms were receiving the identical E-STIM treatment at the same time; however, one arm also had BFR applied periodically throughout the treatment.

Ratings of discomfort

Ratings of discomfort were assessed using the Borg CR10+ scale (Borg, 1990). Individuals were asked to rate their perceived discomfort as they flexed and extended their elbow through the full range of motion. The scale ranged from 0, which represents no discomfort, up to 10 which represents the worst discomfort the individual has ever felt before. Individuals were informed they could provide a rating that exceeded 10 if they perceived more discomfort then they have ever felt before.

Maximal isometric strength

The isometric strength test was performed with the individual in the standing position with their forearm supinated and elbow flexed at a 90° angle. The wrists of the individuals (just proximal to the hand) were strapped to a force transducer (SM-250, Interface, Inc., Scottsdale, AZ, USA) that was secured on a height-adjustable metal stand. The participants were instructed to pull up as hard as they could by trying to lift their forearm by flexing their elbow. Individuals were provided verbal encouragement and visual feedback was given during the test that lasted 5 seconds. The maximum amount of force the individual could produce was recorded as their maximal isometric strength.

Muscle thickness

Ultrasound measurements (Logiq P9, General Electric, Fairfield, CT, USA) were taken at 70% the distance between the acromion process and lateral epicondyle of the anterior upper arm. The probe was coated with gel and held transversely against the skin. The distance between the muscle-fat interface and muscle-bone interface was recorded as the individuals muscle thickness. Two images were taken at each location and averaged together.

Statistics

All statistics were computed using JASP version 0.16 (JASP Team; Netherlands). To compare changes in isometric strength and muscle thickness a 2 (condition: E-stim and E-stim + BFR) × 3 (time: pre, 24 hours, 48 hours) Bayesian repeated measures ANOVA was computed. Since all individuals reported no discomfort on their pre-visit, a Bayesian 2 (condition) × 2 (time) repeated measures ANOVA was computed to assess discomfort without inclusion of the pre-test measure. Changes in discomfort were examined using nonparametric statistics. All Bayesian statistics were computed using uninformed priors, and interactions and main effects were assessed using Bayes Factors (BF_{10}). A BF_{10} of >3 or >10 was used as moderate and strong support for the alternate hypothesis, respectively. A BF_{10} of <0.33 or <0.10 was

used as moderate and strong support for the null, respectively (van Doorn et al., 2021). Post-hoc comparisons were adjusted using Holm's procedure (Holm, 1979). Frequentist statistics were also computed and *p*-values are included along with the BF_{10}.

Results

The results are expressed as means (standard deviations). A total of 18 individuals completed the study (9 females, 9 males). Individuals were 21 (1) years of age, 170.4 (9.9) cm tall, and had a body mass of 77.7 (20.4) kg. The load used for the damaging exercise bouts were 13.1 (4.7) kg and 12.9 (4.6) kg for the E-STIM and E-STIM+BFR arms, respectively. During the damaging exercise bout, individuals complete 50 (16) and 54 (22) repetitions for each the E-STIM and E-STIM+BFR arms. The intensity required to observe a twitch were also similar betweenthe E-STIM and E-STIM+BFR arms on the first [E-STIM: 23.8 (6.5) mA, E-STIM+BFR: 25.0 (4.7) mA] and second [E-STIM: 24.7 (7.9) mA, E-STIM+BFR: 24.3 (6.2) mA] treatment sessions. The mean pressure applied during the E-STIM+BFR protocol was 158 (32) mmHg.

Discomfort

When comparing the E-STIM and E-STIM+BFR protocols, there was no interaction ($BF_{10} = 0.279$; $p = 0.500$), but there was a main effect of time ($BF_{10} = 241.996$; $p = 0.002$) indicating that discomfort increased from the 24 to 48-hour time points (Figure 2). The results were unchanged when computing parametric and nonparametric statistics.

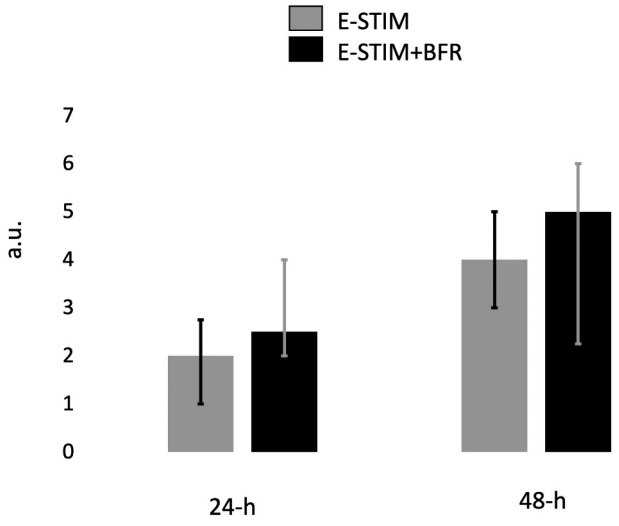

Figure 2. Discomfort. Ratings were given at the 24-hour and 48-hour post-exercise time point. * indicates a difference from the other groups at the 48-hour time point. Values are presented as medians and 95% credible intervals.

Isometric strength

There was no interaction ($BF_{10} = 0.163$; $p = 0.522$) or main effect of condition ($BF_{10} = 0.208$; $p = 0.737$) for changes in isometric strength. There was a main effect of time ($BF_{10} = 10{,}289.894$; $p = 0.001$) indicating that strength was reduced at both the 24-hour ($BF_{10} = 759.153$; $p = 0.004$) and 48-hour ($BF_{10} = 74.352$; $p = 0.003$) time points relative to the pre-test value, but no differences were noted between the 24-hour and 48-hour time points ($BF_{10} = 0.200$; $p = 0.78$; Figure 3).

Muscle thickness

There was no interaction ($BF_{10} = 0.154$; $p = 0.785$) nor was there a main effect of condition ($BF_{10} = 0.388$; $p = 0.260$), but there was a main effect of time ($BF_{10} = 88.476$; $p = 0.012$). Muscle thickness was greater at the 48-hour time point when compared to both the pre-test ($BF_{10} = 15.788$; $p = 0.010$) and 24-hour ($BF_{10} = 10.914$; $p = 0.134$) time points (Figure 4).

Discussion

The primary finding of the present study details that the effectiveness of E-STIM for limiting symptoms associated with muscle damage was not enhanced if performed under BFR. We are confident that our eccentric exercise protocol successfully induced muscle damage as evident by the increased levels of discomfort, muscle swelling, and decrements in isometric strength present. The causes of muscle damage resulting from eccentric exercise are likely multi-factorial. Decrements in strength are thought to occur due to the disorganization of sarcomeres and alterations in calcium kinetics (i.e. calcium release and/or calcium sensitivity) (Morgan & Allen, 1999) which limits the number of cross-bridges and reduces force generation. Muscle soreness is thought to result from the production of neurotrophic factors that induce inflammation and activate nociceptors (Peake et al., 2017). The effectiveness of BFR and/or E-STIM as a means of attenuating the

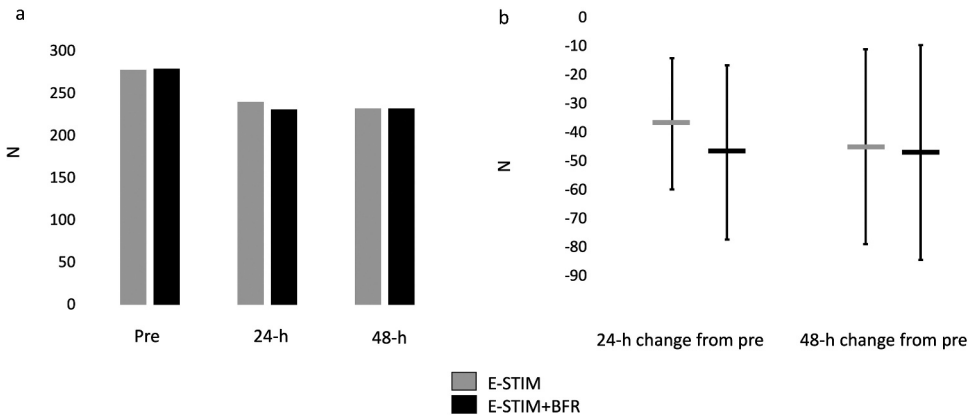

Figure 3. Isometric Strength. a) absolute values at each time point b) mean changes from baseline and the 95% credible interval of this change. Isometric strength values decreased across time with no difference between the treatment conditions.

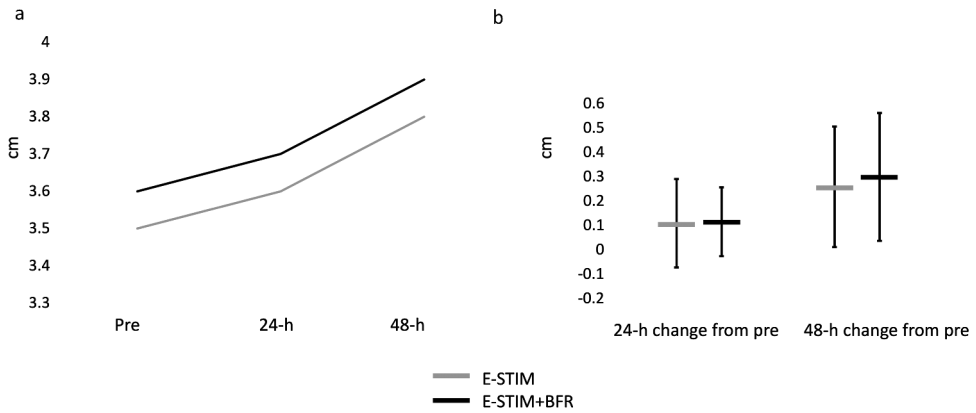

Figure 4. Muscle Thickness. Muscle thickness increased across time with no difference between treatment conditions. a) absolute values at each time point; b) mean changes from baseline and 95% credible intervals of this change.

onset of muscle damage would be dependent upon either reducing or offsetting the factors that initiate muscle damage.

Several studies have found that intermittent bouts of BFR applied immediately after damaging exercise were capable of reducing strength loss and muscle soreness (Beaven et al., 2012; Page et al., 2017). The authors hypothesized that this was due to a reduced inflammatory response relating to an increase in nitric oxide or down-regulation of leukocytes (Page et al., 2017). Indeed, these results are supported by a recent meta-analysis demonstrating that BFR applied in the absence of exercise may limit muscle damage and accelerate recovery (Arriel et al., 2020). The authors also postulate that the benefits of BFR applied post-exercise is likely related to an increased nitric oxide release and subsequent reduction in reactive oxygen species (Arriel et al., 2020). While these studies incorporated BFR in the absence of exercise, one study applying BFR in conjunction with unweighted elbow flexion exercises found no attenuation of muscle damage when applied immediately following damaging exercise (Wong et al., 2020). Our results would support those of Wong and colleagues by questioning the effectiveness of BFR as a means of attenuating the magnitude of muscle damage, particularly when used in conjunction with E-STIM.

The application of E-STIM was previously shown to attenuate creatine kinase levels when applied after damaging exercise, but was unable to limit muscle soreness or the magnitude of strength loss (Vanderthommen et al., 2007). Our findings provide further insight that any potential efficacy of E-STIM for attenuating symptoms of muscle damage is unlikely to be enhanced if performed under BFR. It would seem reasonable to hypothesize that we did not observe any added benefit of applying BFR to E-STIM, because the E-STIM protocol itself may have resulted in a similar redundant physiologic response capable of mitigating muscle damage (e.g. increased blood flow, reduced inflammatory response). It is also possible that neither the E-STIM protocol nor the BFR protocol were sufficient to alleviate symptoms of muscle damage.

While not observed in the present study, there still may be utility of applying BFR to E-STIM protocols depending on when the treatment is administered. In the present study, we elected to administer the treatment protocol immediately post and 24-hours

post damaging exercise. Previous research has demonstrated that BFR exercise (Sieljacks et al., 2016) and E-STIM (Black & McCully, 2008) can reduce the magnitude of future muscle damage (i.e. the repeated bout effect), suggesting that administering this combined intervention before the damaging exercise may have been effective. Even so, the effectiveness of these protocols at alleviating future muscle damage would likely be related to the magnitude of muscle damage induced by the "treatment protocol". This limits the utility of this approach as the purpose of the present study was to prevent muscle damage as opposed to inducing it with the treatment protocol. Other studies would suggest that both BFR (Song et al., 2021) and E-STIM interventions (Babault et al., 2011) may be effective at acutely alleviating muscle soreness as opposed to limiting or preventing the onset of muscle damage. Thus, the protocols could have been administered after muscle damage presented itself in attempt to treat as opposed to prevent muscle damage.

There were several identifiable limitations within the present study. First, we did not have a true control condition, nor did we have a condition incorporating BFR in the absence of E-STIM. We elected to compare E-STIM and E-STIM+BFR using a within subject design to allow for a direct comparison of muscle soreness, strength, and swelling at the same timepoints following the damaging exercise. We felt this was the best design to answer the question as to whether BFR could augment the benefits of E-STIM. Furthermore, we elected to use the elbow flexor muscles as opposed to a larger, more active, muscle group such as the quadriceps. This was done to minimize the influence of extraneous activities performed outside of the research study (e.g. walking, climbing stairs, etc.). Lastly, we used a low frequency E-STIM and an intermittent BFR protocol which were performed immediately post and 24-hours post damaging exercise. Thus, the results of this study may be dependent upon the specific protocols chosen and the temporality of when the treatments were administered relative to the damaging exercise bout.

Conclusion

While BFR itself may have utility for preventing the onset of muscle damage, it does not appear to have any additive effect when combined with low frequency E-STIM. Future studies may wish to examine alternative E-STIM and/or BFR protocols aimed at preventing the onset of muscle damage.

Disclosure statement

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Ethical publication statement

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Availability of data

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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